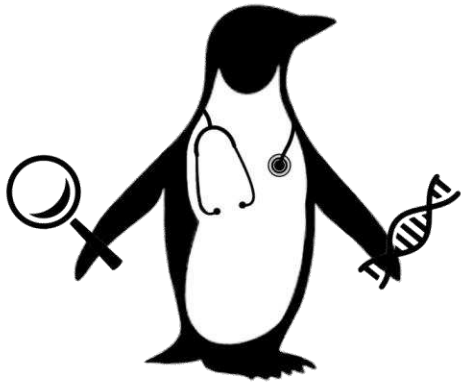
The PENGQUIN Odyssey:

A Quest for Query-able Semantic Genomic Data for Clinical Practice



Elias Crum

SIB Vital-IT Group Meeting
April 4th, 2025

SIB Research Stay: 1 March – 30 April

A Quick Thank You!

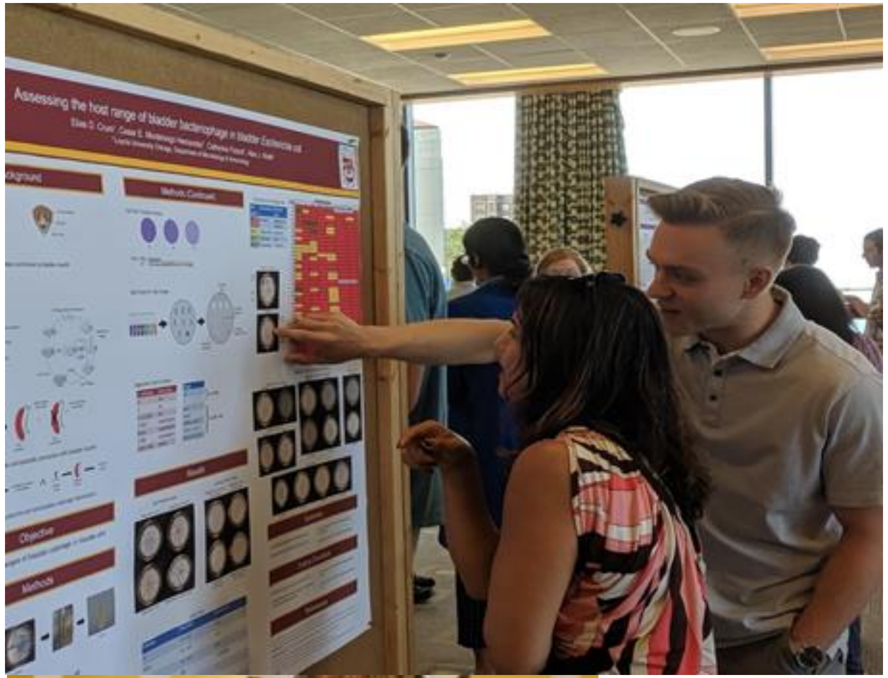
Tarcisio Mendes

Ana-Claudia Sima

Jerven Bolleman

Marco Pagni

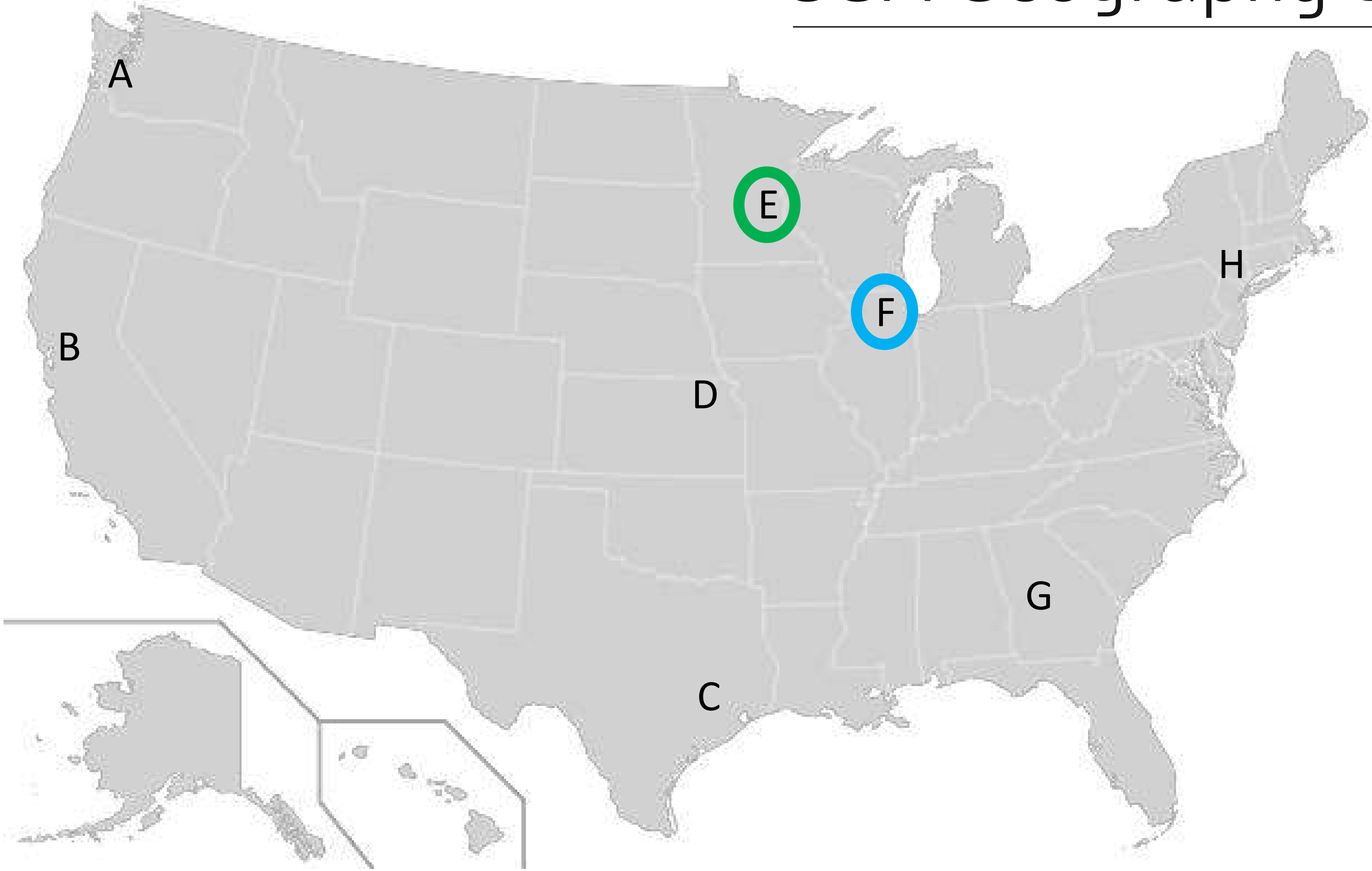
An Intro to Elias

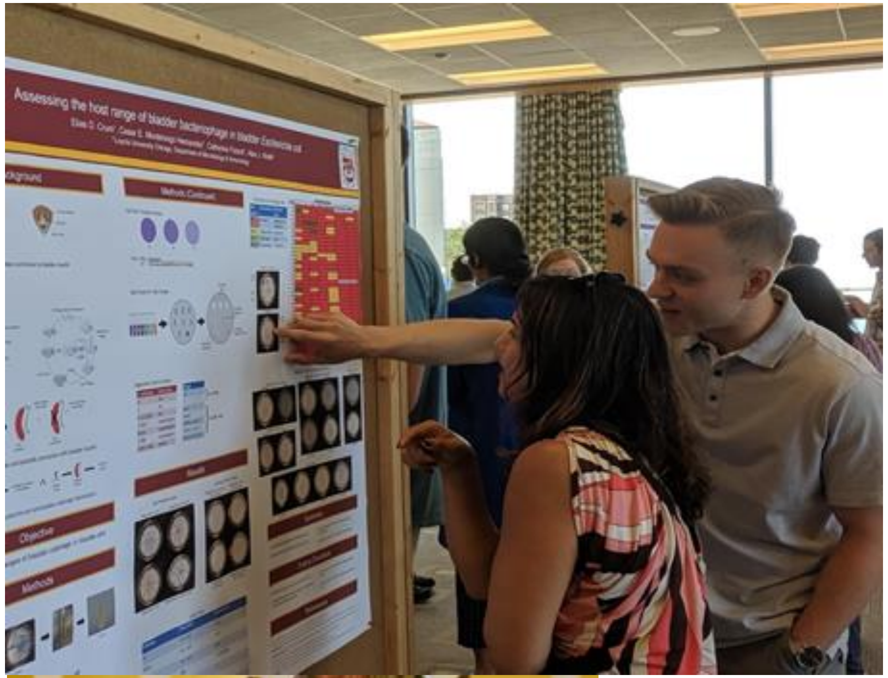


My Journey:

BSc + MSc - Bioinformatics

USA Geography Lesson





An Intro to Elias

My Journey:

BSc + MSc - Bioinformatics

Prokaryotic Genomics

How can we increase **genomics** use in **medicine**?

PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice

Genomic Data → Clinical Care

Semantic, Linked Data

Query Capabilities

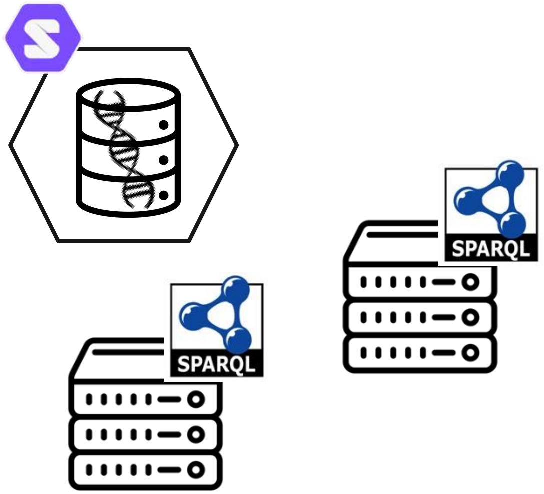
Framework + App

Advisors: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2]

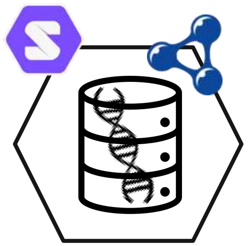
1 = vito 2 = Ghent University

PENGQUIN Goals

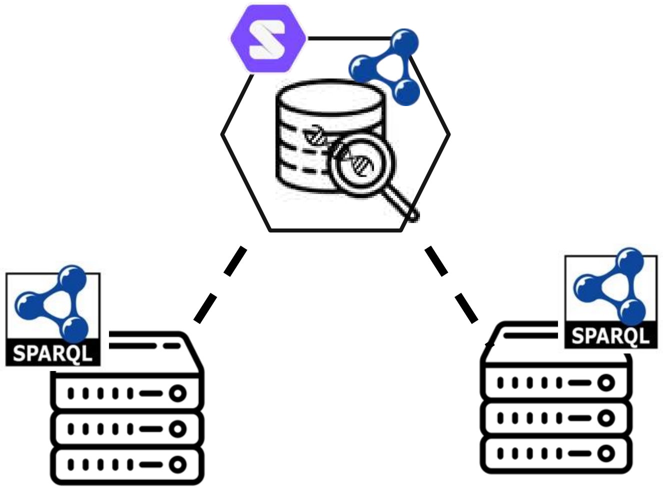
Data Storage



Genomic Data Representation



Question Answering



Framework
& web application

PENGQUIN Goals

Improve genomic data USAGE and SCALABILITY

Improve genomic data store accessibility (while maintaining privacy)

Represent genomic data semantically as Linked Data

Query genomic data and linkages

Beacon

Connect to existing initiatives / applications

Proof-of-concept

Genome Pods

Develop a framework and web application

Talk Outline

Decentralized Data Storage *(CHIST-ERA TRIPLE Project)*

- Introduction to Solid Pods
- Introduction to TRIPLE Project
- The Solid Cockpit web application (and maybe a use-case example if we have time)

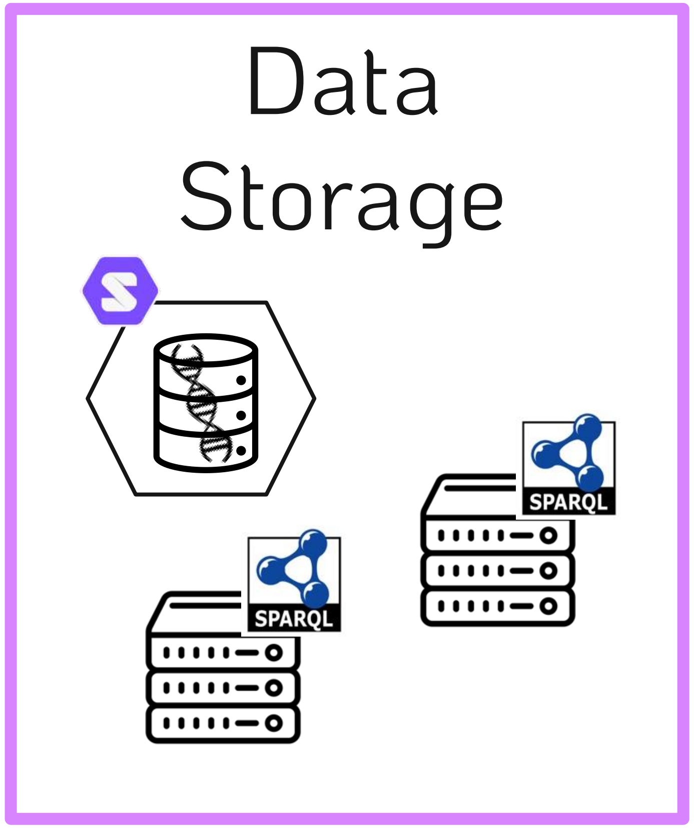
Genomic Variant Data Semantification *(Semantic VCF)*

- Why Semantic VCF Data?
- Existing vocabularies/tools

Federated Querying *(Real World Federated Query Survey)*

- Motivation + Challenges
- Progress and preliminary findings
- Future aims

The First Act – Private Data Storage



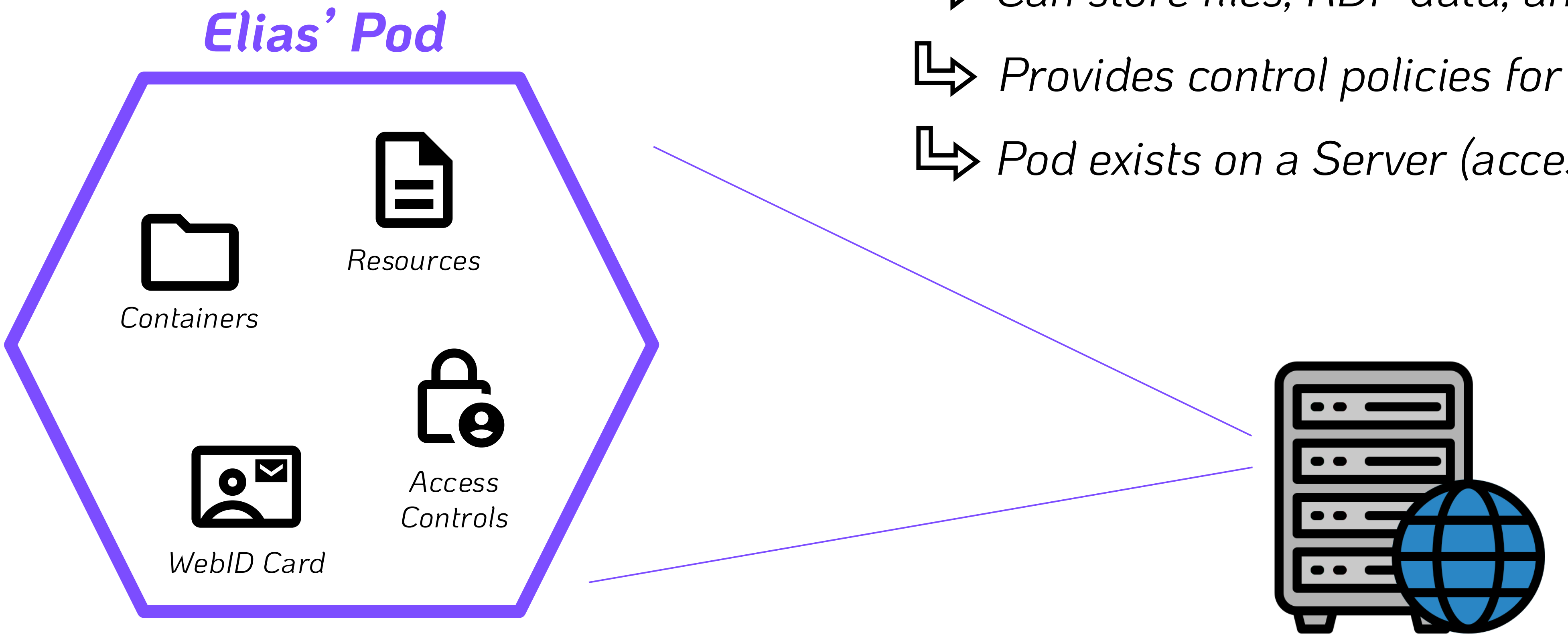
CHIST-ERA TRIPLE Project

Address current **challenges** of using **RDF resources**
to bring the **utility** of these resources to a **broader group of researchers**

What is a Solid Pod?

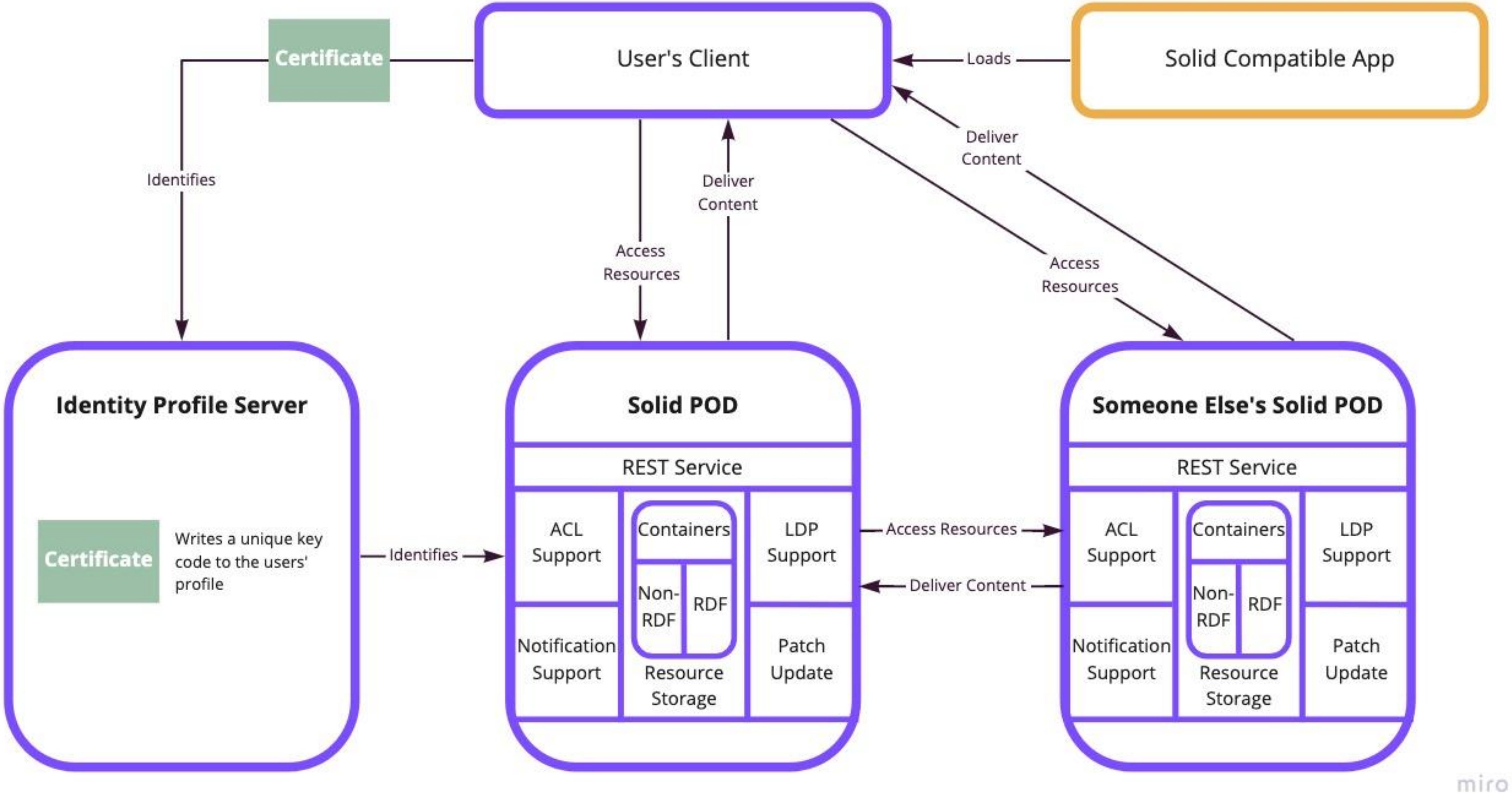
Secure, Decentralized Data Storage Specification

- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*
- ↳ *Pod exists on a Server (accessible to the Web)*



https://solidproject.org/about

Solid's Architecture



https://solid.georgetown.domains/technical-component/

Solid Pod vs SPARQL Endpoint?

Solid Pod

+ decentralized storage strategy
+ can store both file data and RDF data
+ granular privacy policy capabilities

- native querying capabilities limited
- not designed for large-scale data
- no computation attached to data store

SPARQL Endpoint

- centralized storage strategy
- triple store (only RDF)
- limited privacy capabilities

+ designed for high-performance querying
+ excels with large datasets
+ computation directly attached to data store

How to query a Solid Pod?

A Query Engine

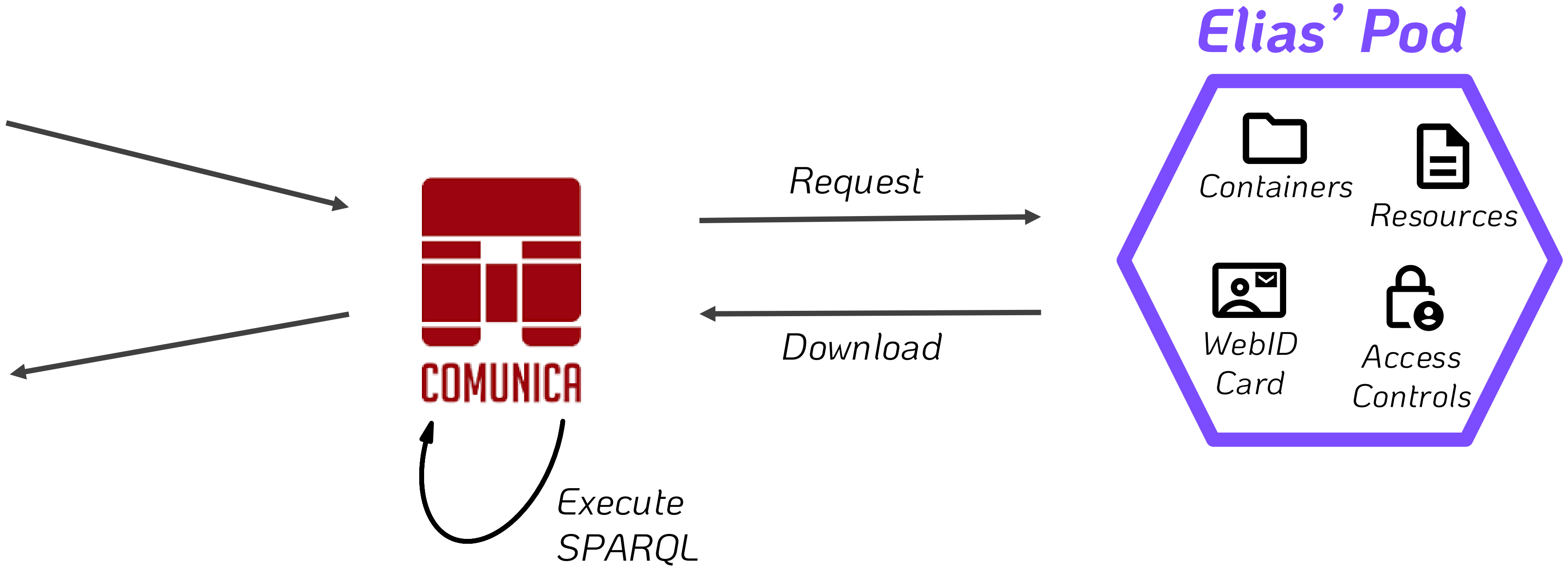
- Query computation client-side
- Allows for SPARQL query execution “over” Solid Pod API
- Linked Data traversal + heterogeneous result joining capabilities

SPARQL Query

```
SELECT DISTINCT ?p {
    ?s ?p ?o
} LIMIT 10
```

+ Source:
Elias’ Pod

Results Binding Stream
SPARQL+JSON



https://comunica.dev/about

Solid Pod Implementations?

BBC MyPDS App

Flemish Data Utility Company

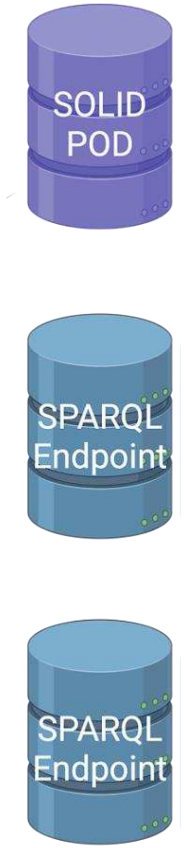
health data vaults

Flemish Smart Dataspace

Solid Lab Use-cases

CHIST-ERA TRIPLE Project

Swiss Institute of Bioinformatics



Standardized documentation and resource statistics

My Contributions

Solid Cockpit

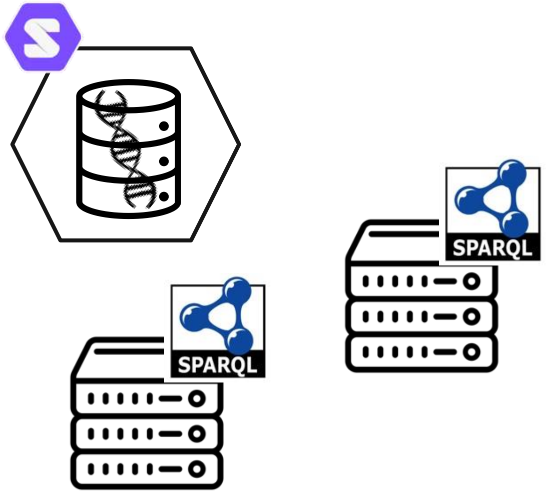
A Web Application to:

knowledgeonwebscale.github.io/solid-cockpit/

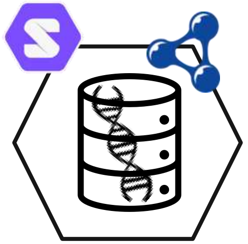
- Upload Data to a Solid Pod
- Browse contents of a Solid Pod
- Edit Privacy of (and share) data in a Solid Pod
- Query Data in Solid Pods + SPARQL Endpoints
    - ↳ Implement a query cache for storing queries and results

The Second Act – Semantifying Genomic Variant Data

Data Storage



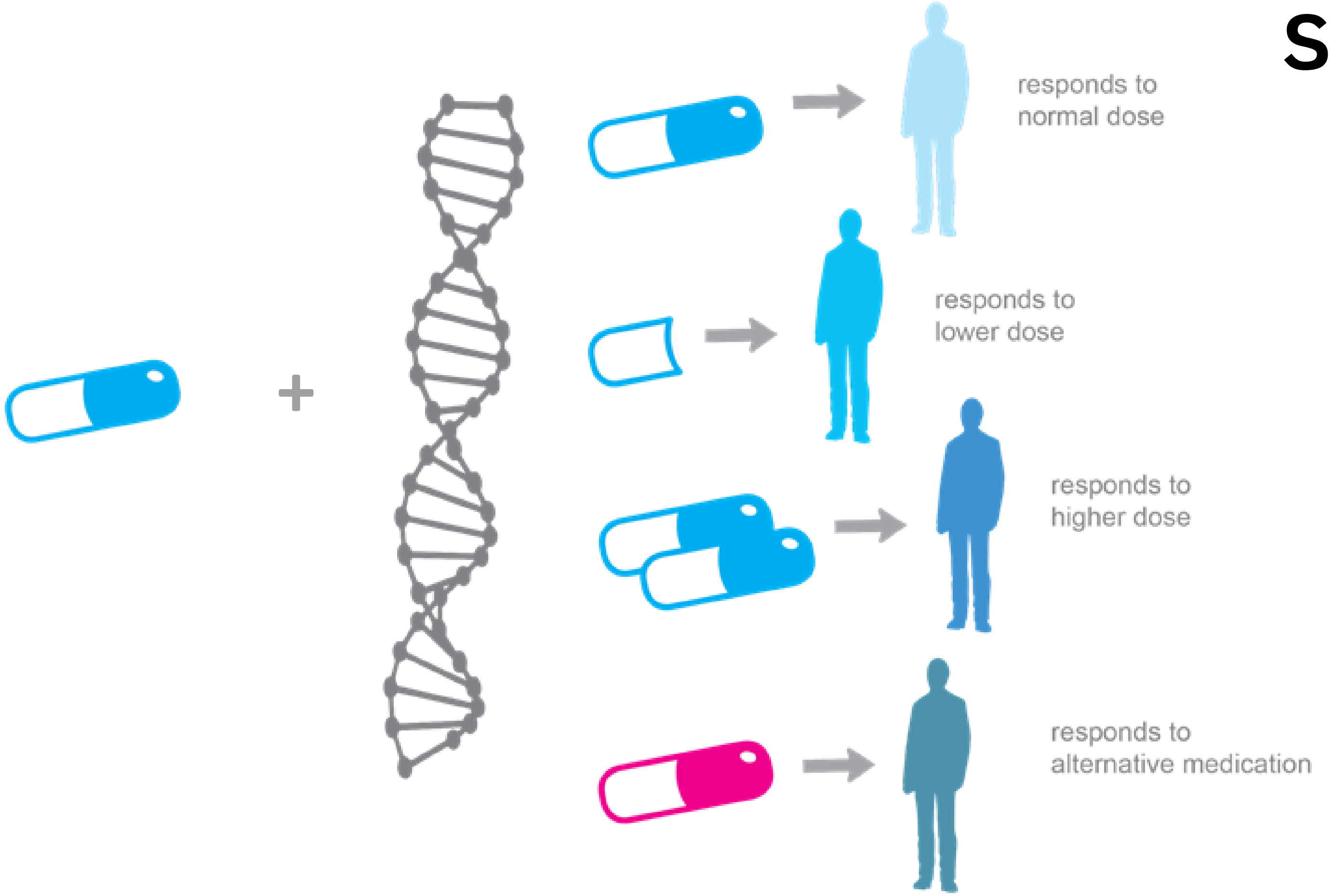
Genomic Data Representation



Representing Genomic Variant Data Semantically

Genomic Data, Health Care, and Semantics?

Pharmacogenomics Use-case (E.g. Warfarin):



https://alphagenomix.com/empowering-personalized-medicine/

Semantic Genomic Variant Data?

Research:

⮡ **F.A.I.R.** –
Findable. Accessible.
Interoperable. Reproducible.

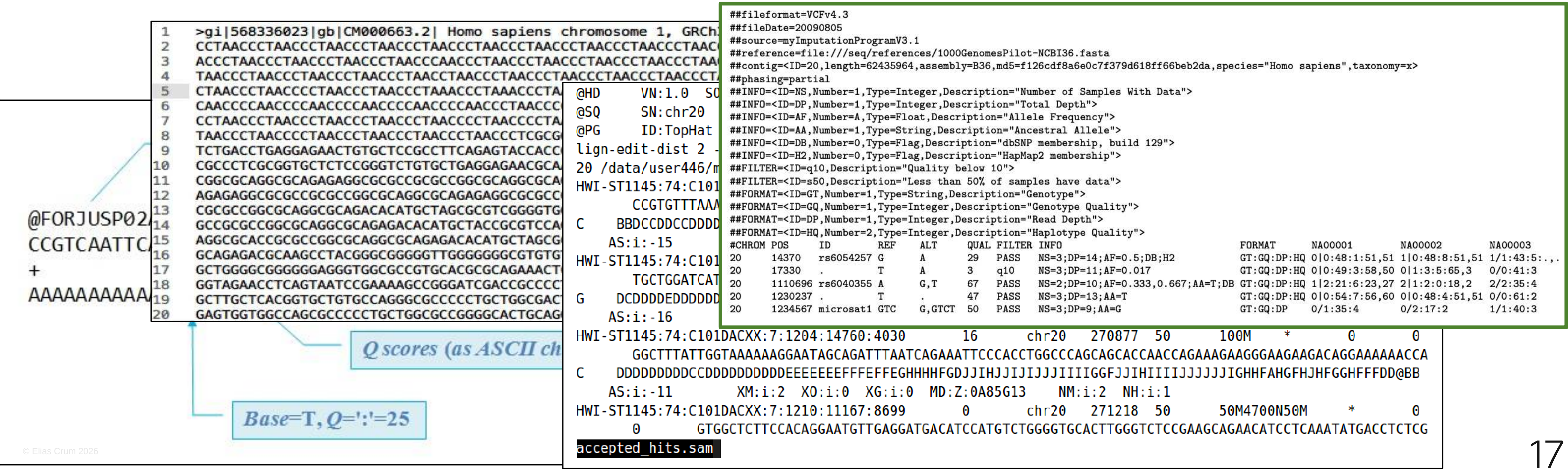
Clinic:

⮡ Data Linking
Data Provenance
Data Interoperability

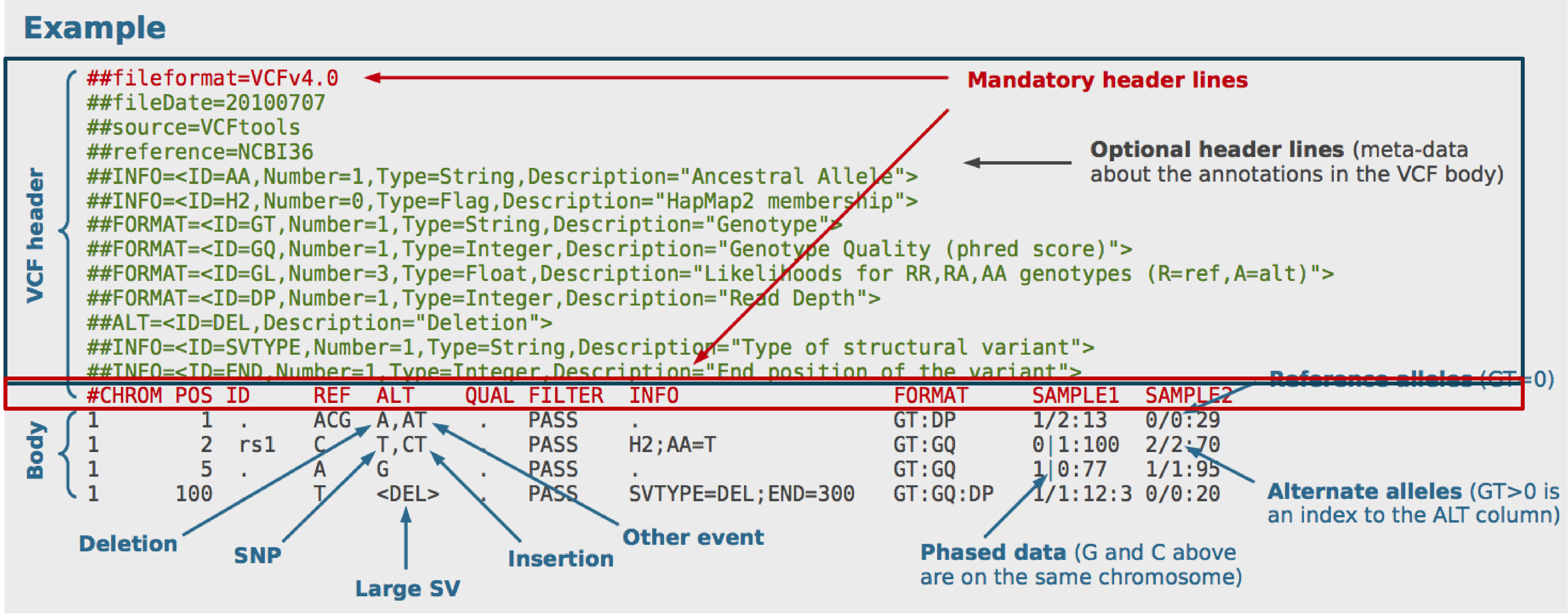
Semantic VCF – *Sequencing Data Formats Intro*

FASTQ ✚ **FASTA** → **SAM/BAM** → **VCF**

FASTQ	FASTA	SAM/BAM	VCF
Raw Sequencing Outputs	Contiguous DNA sequence	FASTQ reads aligned to a FASTA reference	Areas in SAM where sample and reference differ



What does a VCF File Look like?

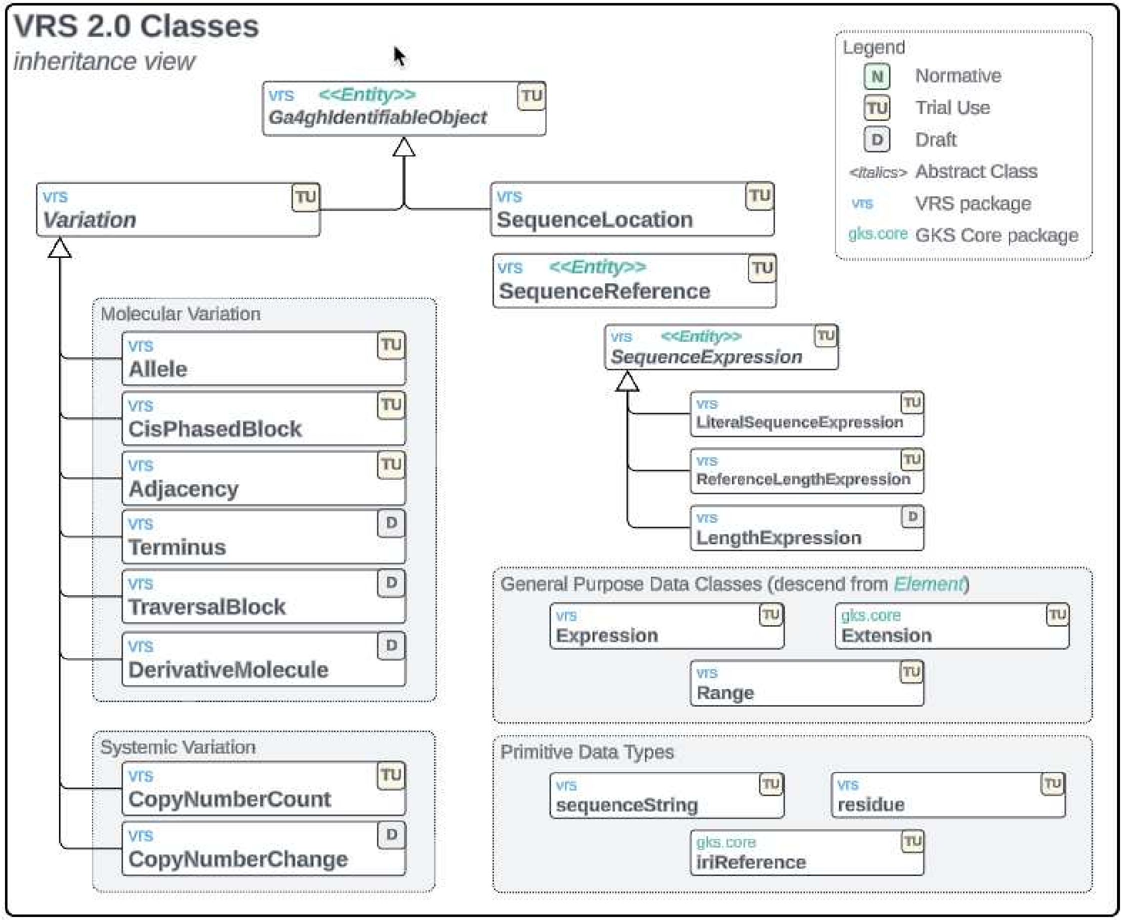


https://davetang.github.io/learning_vcf_file/

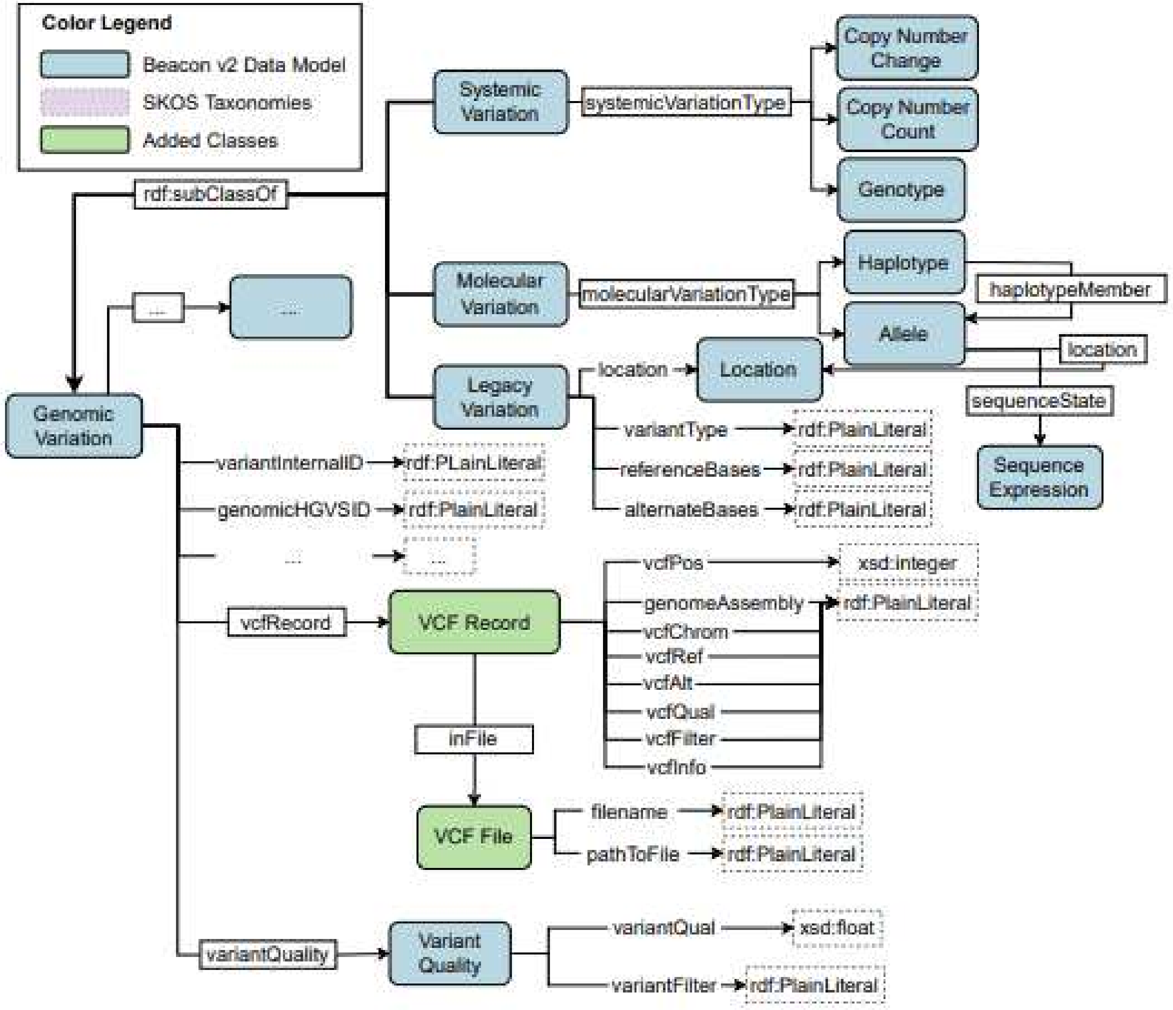
VCF Spec: https://samtools.github.io/hts-specs/VCFv4.2.pdf

Representing VCF Semantically?

GA4GH Variant Schema:



HERO Genomics Ontology:



https://vrs.ga4gh.org/en/2.0.0-ballot.2024-11/appendices/class_diagram.html#classdiagram

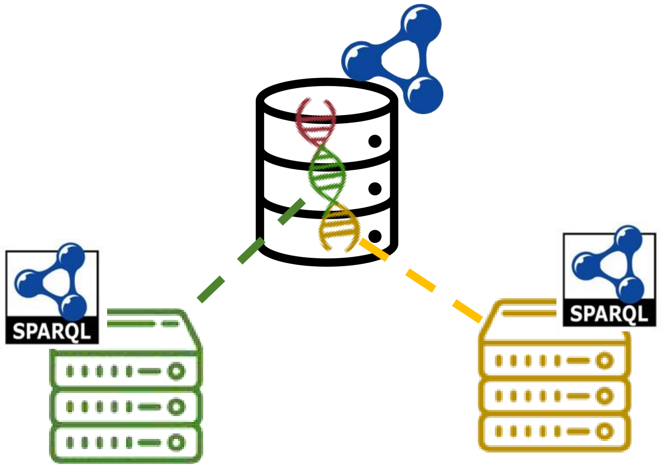
https://www.dei.unipd.it/~silvello/papers/2025_MCS_SWAT4HCLS.pdf

Semantic VCF - *Future Aims*

Construct VCF Knowledge Graph

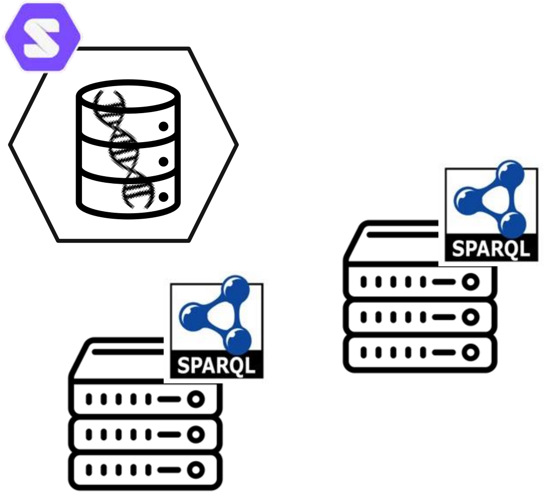
rml.io **OR** RocketRML

Experiment with SPARQL Querying and Data Linking

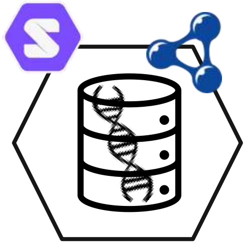


The Third Act – SPARQL Federated Querying

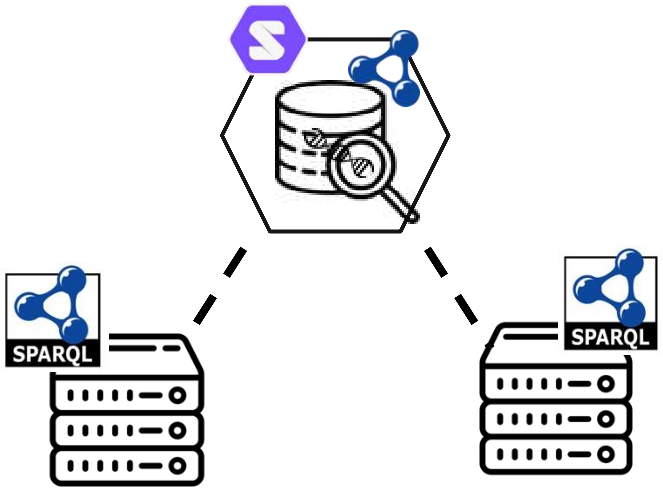
Data Storage



Genomic Data Representation



Question Answering



Real World Federated Query Survey

Swiss Institute of Bioinformatics

GHENT UNIVERSITY

Motivation:

Federated querying for answering questions over **multiple heterogeneous data sources**

Performance is currently ~*variable*~

↳ We want to detail *WHY* performance varies so much && suggestions for improvement

Assess federation algorithms on REAL endpoints

↳ ***FedX*** – uses `ASK` queries to inform source-selection

↳ ***SPLENDID*** – `VoID` descriptions to inform source-selection

Real World Federated Query Survey - *Design*

SPARQL Endpoints:

“Real World” Federated Queries: https://sib-swiss.github.io/sparql-examples/

SPARQL Query Engine: https://comunica.dev/

Example Federated Query

Query: 027-biosodafrontend

Question: (*Bgee* | *UniProt*)

Retrieve the *genes* expressed in the *human pancreas*

and

those *genes*' *annotations*

https://www.bgee.org/sparql/

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX up: <http://purl.uniprot.org/core/>
PREFIX obo: <http://purl.obolibrary.org/obo/>
PREFIX genex: <http://purl.org/genex#>
PREFIX lscr: <http://purl.org/lscr#>
PREFIX orth: <http://purl.org/net/orth#>
SELECT DISTINCT ?geneEns ?uniprot ?annotation_text {
    {
        SELECT ?geneEns {
            ?geneB a orth:Gene .
            ?geneB genex:isExpressedIn ?cond .
            ?cond genex:hasAnatomicalEntity ?anat .
            ?geneB lscr:xrefEnsemblGene ?geneEns .
            ?anat rdfs:label 'pancreas' .
            ?geneB orth:organism ?o .
            ?o obo:RO_0002162 ?taxon2 .
            ?taxon2 up:commonName 'human' .
        } LIMIT 100
    }
    SERVICE <https://sparql.uniprot.org/sparql> {
        ?uniprot rdfs:seeAlso / up:transcribedFrom ?geneEns .
        ?uniprot up:annotation ?annotation .
        ?annotation rdfs:comment ?annotation_text .
    }
}
```

Example obtained from: https://biosoda.expasy.org/build_biosodafrontend/

visualization from: https://sib-swiss.github.io/sparql-editor/

Real World Federated Query Survey – *[Preliminary] Findings 1*

Preliminary Results (endpoints and queries):

Federated querying is not ”easy”

↳ Errors related to SPARQL endpoint availability

Invalid SPARQL endpoint response from https://semantic.eea.europa.eu/sparql (HTTP status 404)

↳ Errors caused by SPARQL endpoint-related errors

Invalid SPARQL endpoint response from https://sparql.orthodb.org/sparql (HTTP status 500)

↳ Errors related to SPARQL endpoint rate-limiting

Invalid SPARQL endpoint response from https://query.wikidata.org/sparql (HTTP status 429): Rate limit exceeded

↳ Errors related to SPARQL endpoint time-outs

Unexpected ""!"" at position 0 in state STOP

Real World Federated Query Survey – *[Preliminary] Findings 2*

Preliminary Results (algorithms):

Federated querying is not "easy"

↳ Errors within Comunica

Actor urn:comunica:default:query-operation/actors#source requires an operation with source annotation.

fetch failed

Could not retrieve https://idsm.elixir-czech.cz/sparql/endpoint/chebi (HTTP status 400)

FATAL ERROR: Ineffective mark-compacts near heap limit Allocation failed - JavaScript heap out of memory

↳ Errors related to algorithms

Actor urn:comunica:default:/actors#rdf-join cannot complete join with Asym-hash actor.

Real World Federated Query Survey – *Future Work*

Future Aims:

Execute 1 month worth of queries

↳ 4 experimental runs per week

Vary time + part of week experiment is performed

Publication goals:

- In-Use Paper
- Journal Paper Extension (benchmark creation)

Algorithmic variation

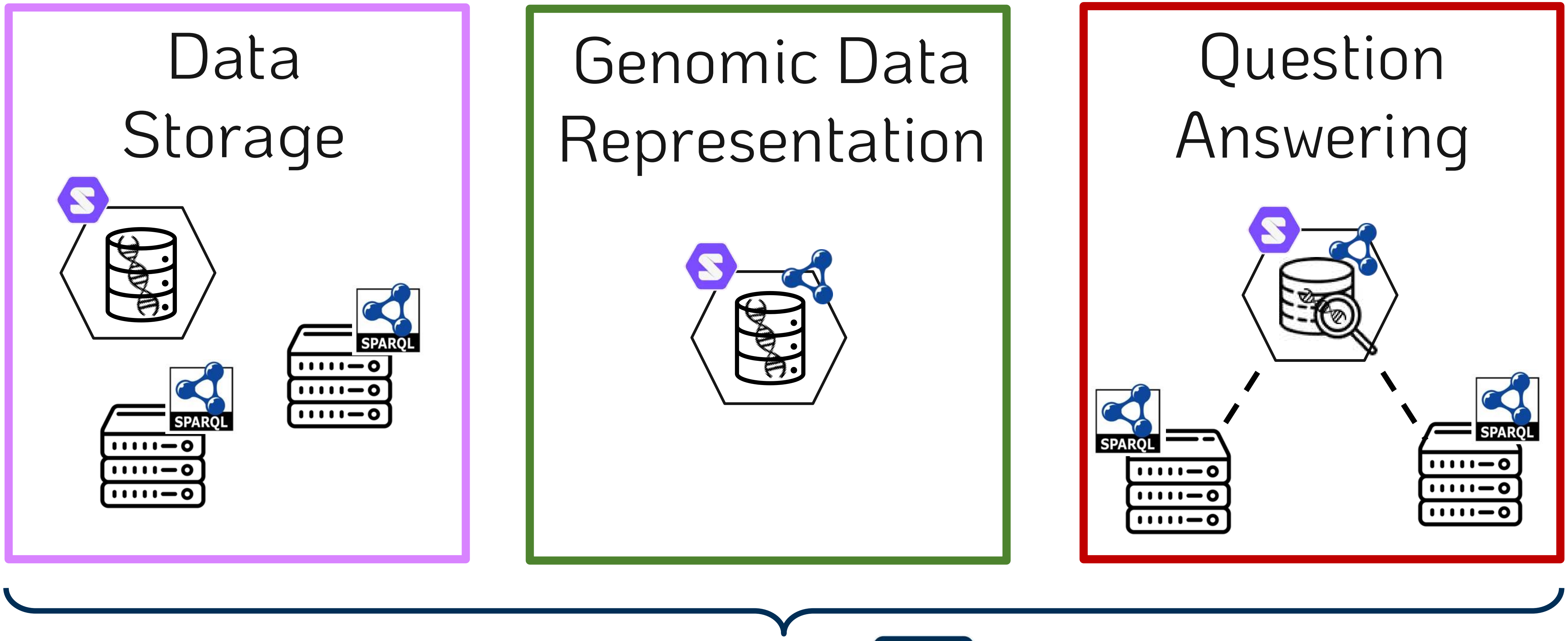
- `VoID` vs `ASK` queries
- Various join algorithm combinations
- Rate-limiting
- `SERVICE` removal

Metrics Assessed

- Total execution time
- Query planning time
- Result set consistency
- # SPARQL requests
- # HTTP requests
- Individual query arrival times

For more detail see: https://docs.google.com/document/d/1U-RtYsJw9raw4U5Qe6BHfSh2VYGnUIkbT9-aSyJionU/edit?usp=sharing

The Finale – Unified System for Storage & Representation & Querying



Framework
& web application

GHENT UNIVERSITY

IDLab INTERNET & DATA LAB

vito

fwo

Thank you!

More PhD specifics

LinkedIn

GitHub

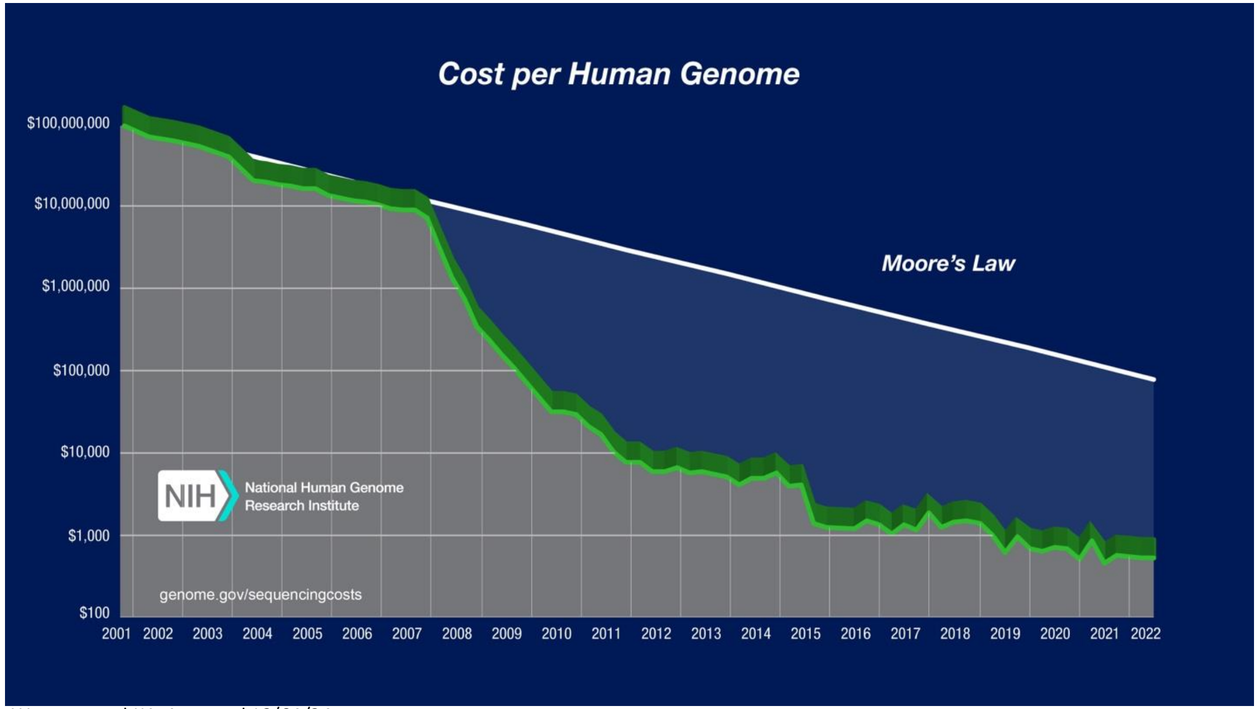
Supplemental Slides

Solid Cockpit

knowledgeonwebscale.github.io/solid-cockpit/

Genome Sequencing in Health Care

Sequence Generation Costs



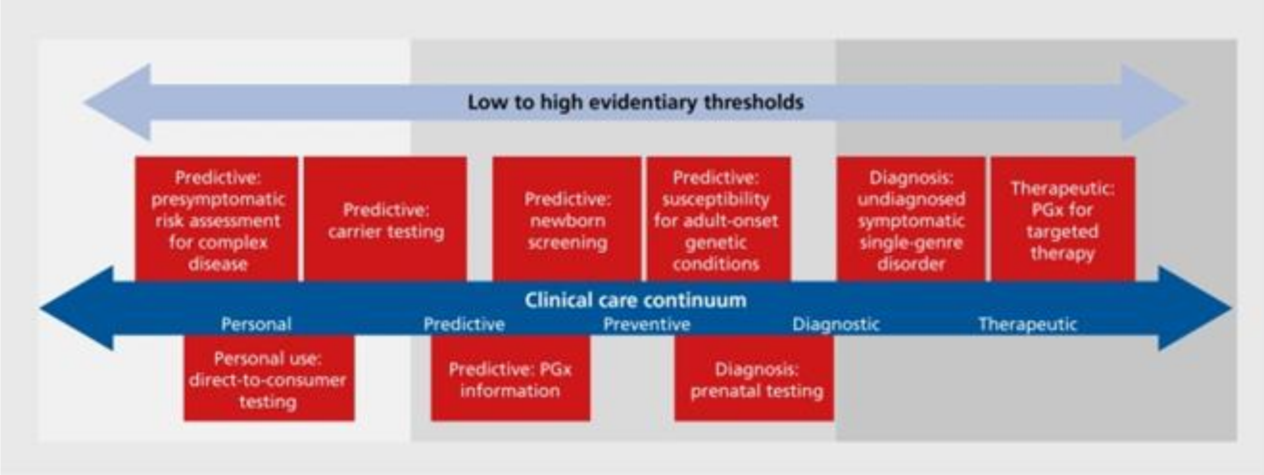
Wetterstrand KA. Accessed 12/04/24.

Currently getting close to $100 per genome[4]

Per-base cost of sequencing has been dropping by half every ~5 months

Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Variant Call Format (VCF)

- tab delimited .txt file

- Rows represent individual mutations in a sequence (when compared to reference)

- Columns store information about that mutation

```
#CHROM  POS ID  REF ALT QUAL    FILTER  INFO    FORMAT
```

```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file://23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM  POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  GEN
chr1    82154   rs4477212   a   .   .   .   .   GT  0/0
chr1    752566  rs3094315   g   A   .   .   .   GT  1/1
chr1    752721  rs3131972   A   G   .   .   .   GT  1/1
chr1    798959  rs11240777  g   .   .   .   .   GT  0/0
chr1    800007  rs6681049   T   C   .   .   .   GT  1/1
chr1    838555  rs4970383   c   .   .   .   .   GT  0/0
chr1    846808  rs4475691   C   .   .   .   .   GT  0/0
chr1    854250  rs7537756   A   .   .   .   .   GT  0/0
chr1    861808  rs13302982  A   G   .   .   .   GT  1/1
chr1    873558  rs1110052   G   T   .   .   .   GT  1/1
chr1    882033  rs2272756   G   A   .   .   .   GT  0/1
chr1    888659  rs3748597   T   C   .   .   .   GT  1/1
chr1    891945  rs13303106  A   G   .   .   .   GT  0/1
```

Personal Genome Sequence Data

Human genome data is big…

~3.2 BILLION base pairs ----> ~1 GB

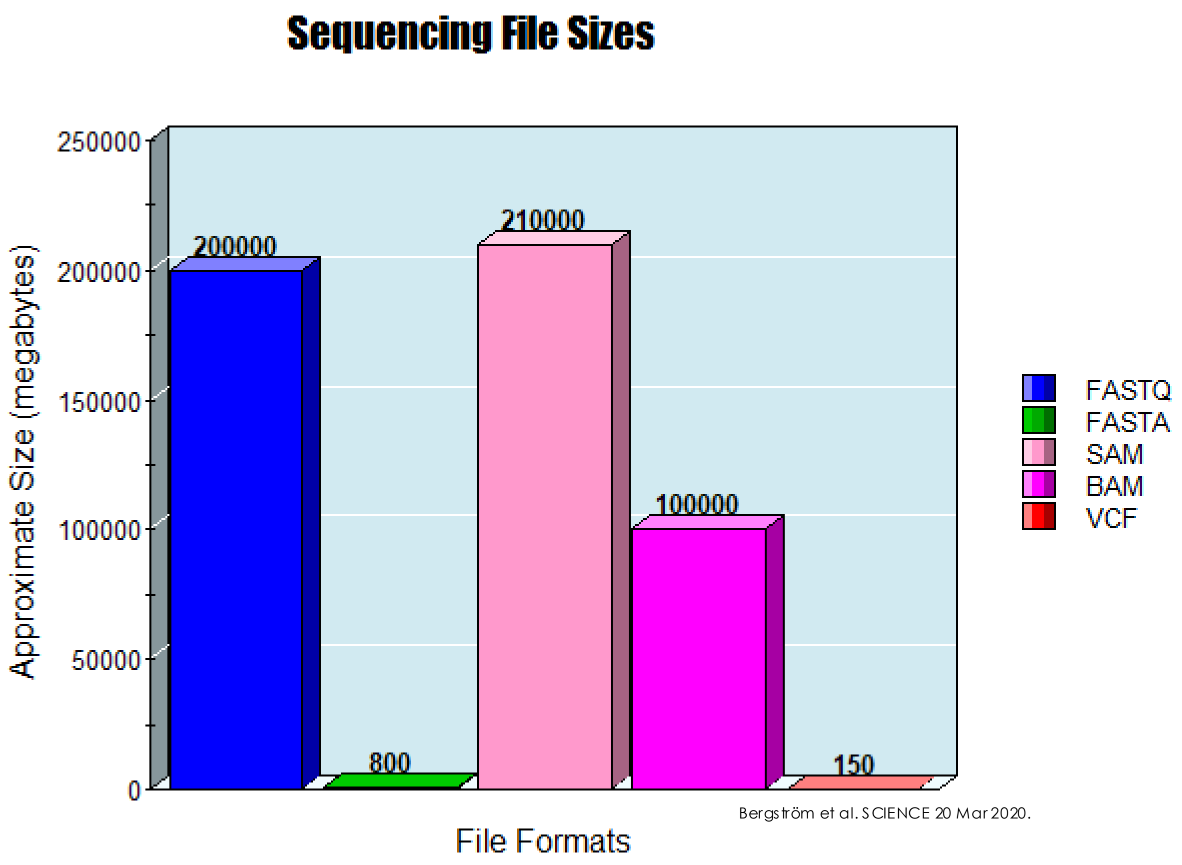
Sequencing output file ----> ~200 GB

Variant Call Format (***VCF***) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

VCF file semantic representation

- **~ >27 MILLION** Triples when represented as RDF
- Ontology for conversion
- Header Dictionary Triples (for compression)



Bergström et al. SCIENCE 20 Mar 2020.

Scaling Genome Data Usage

Research genomic data sharing initiatives:

FHIR

Beacon

OHDSI

OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces → Not much...

Experimental VCF Data

Primary dataset from publicly available repositories (>100 genomes)

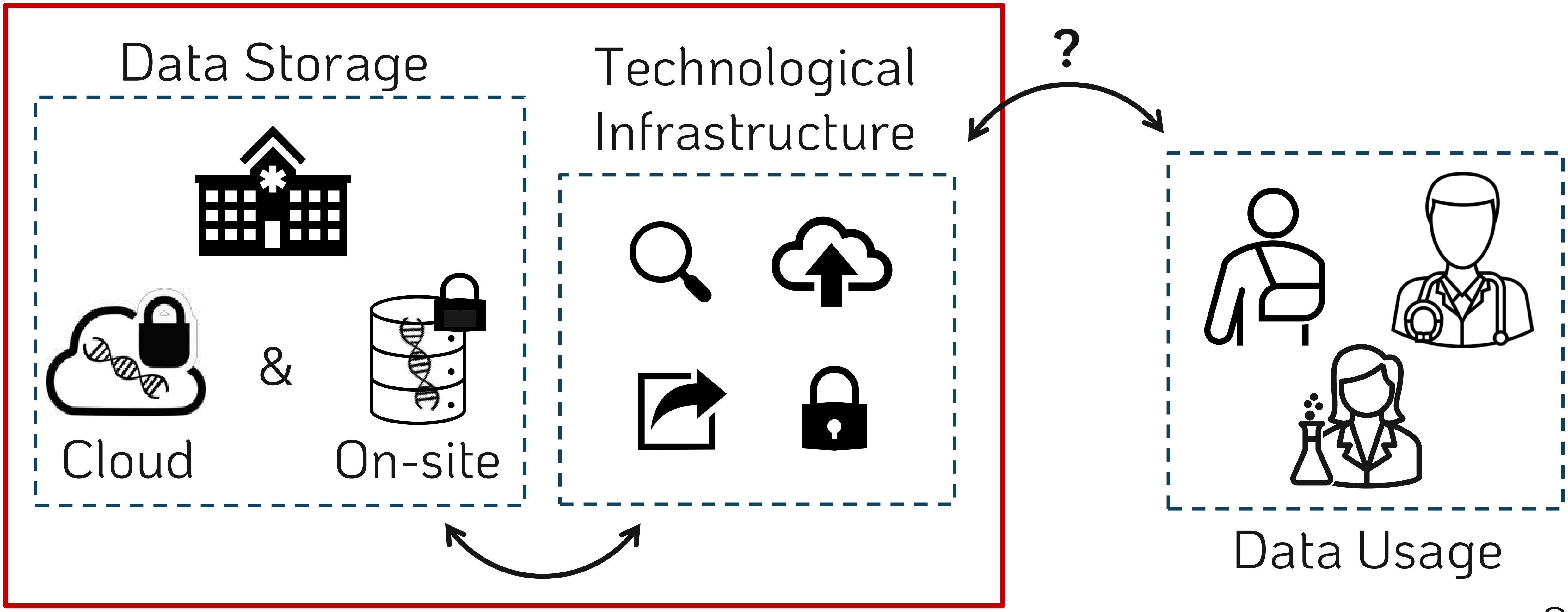
- Personal Genome Project (https://my.pgp-hms.org/public_genetic_data)
- NIST's Genome in a Bottle public dataset (https://data.nist.gov/od/id/mds2-2336)
- Illumina Platinum Genomes repository (https://emea.illumina.com/platinumgenomes.html)

If more are needed (>1000 [research grade] genomes):

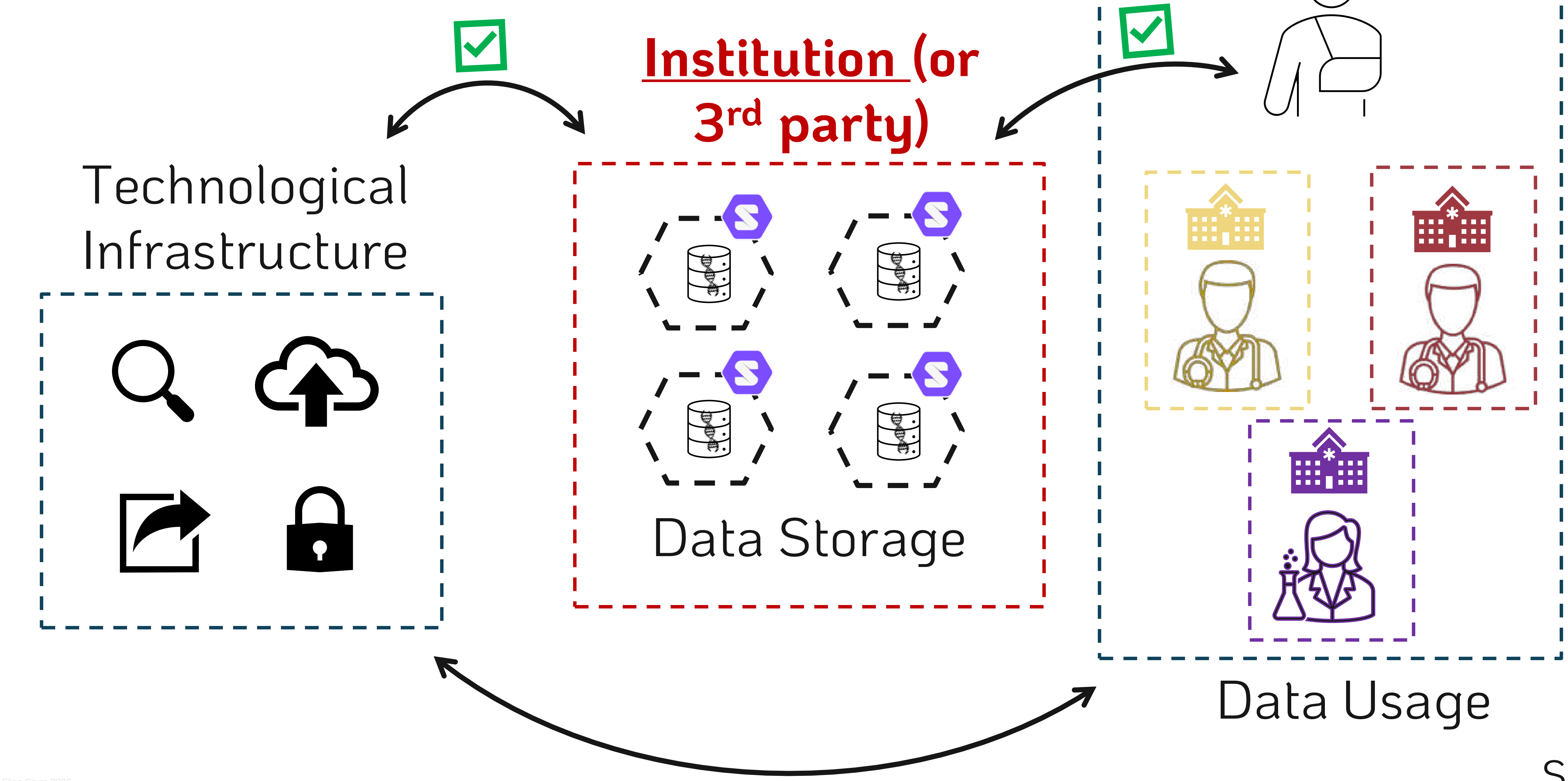
- 1000 genomes project (https://www.internationalgenome.org/)
- NCBI's ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/)

Semantic Genome Data?

Institution-centric



A [Hybrid] Solid Solution



Solid Technology

- Does not solve ALL problems, but offers an environment where improvements are possible

  - ✓ Easier sharing (web-facilitated)
  - ✓ Granular privacy controls
  - ✓ Direct patient data access
  - ✓ Data representation flexibility (RDF)
  - ✓ Possibility of data-linking and querying

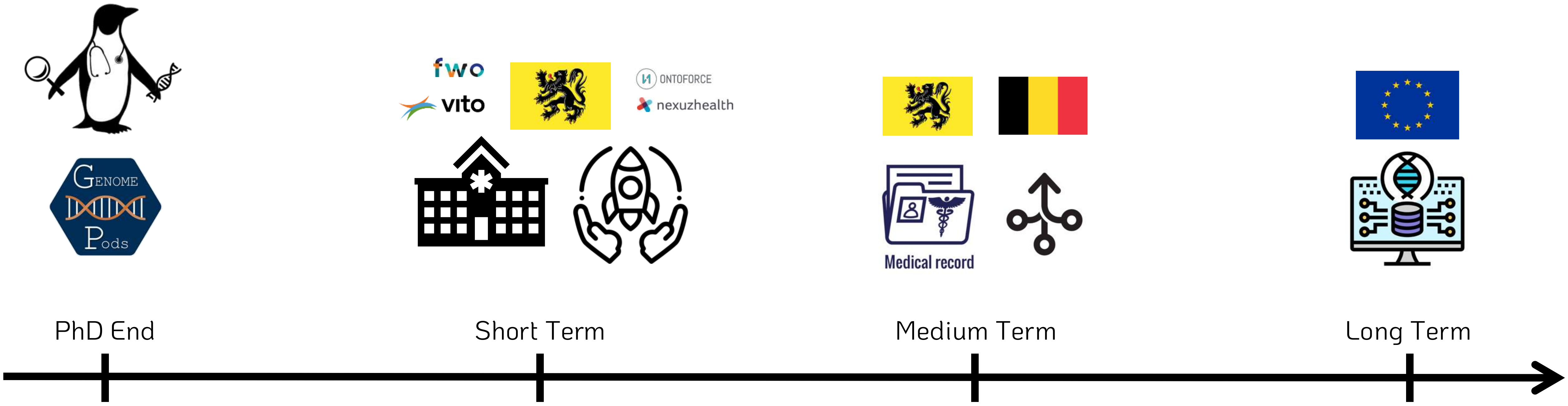
https://solidproject.org/TR/protocol

Project Goals

Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

Envisioned Product Evolution



PhD
Deliverable

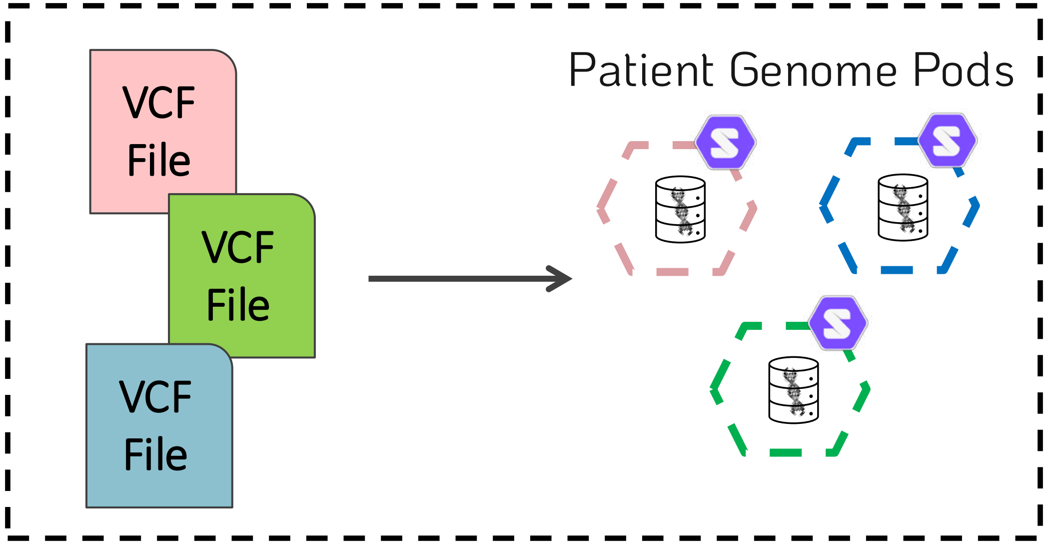
Start-up / Product
development
+
Hospital implementation

Integrate with EHR
+
Increase application
domains

Increase 3rd party
application support
+
Scale to Europe and
beyond

Workflow Overview

Data Storage



Data Storage

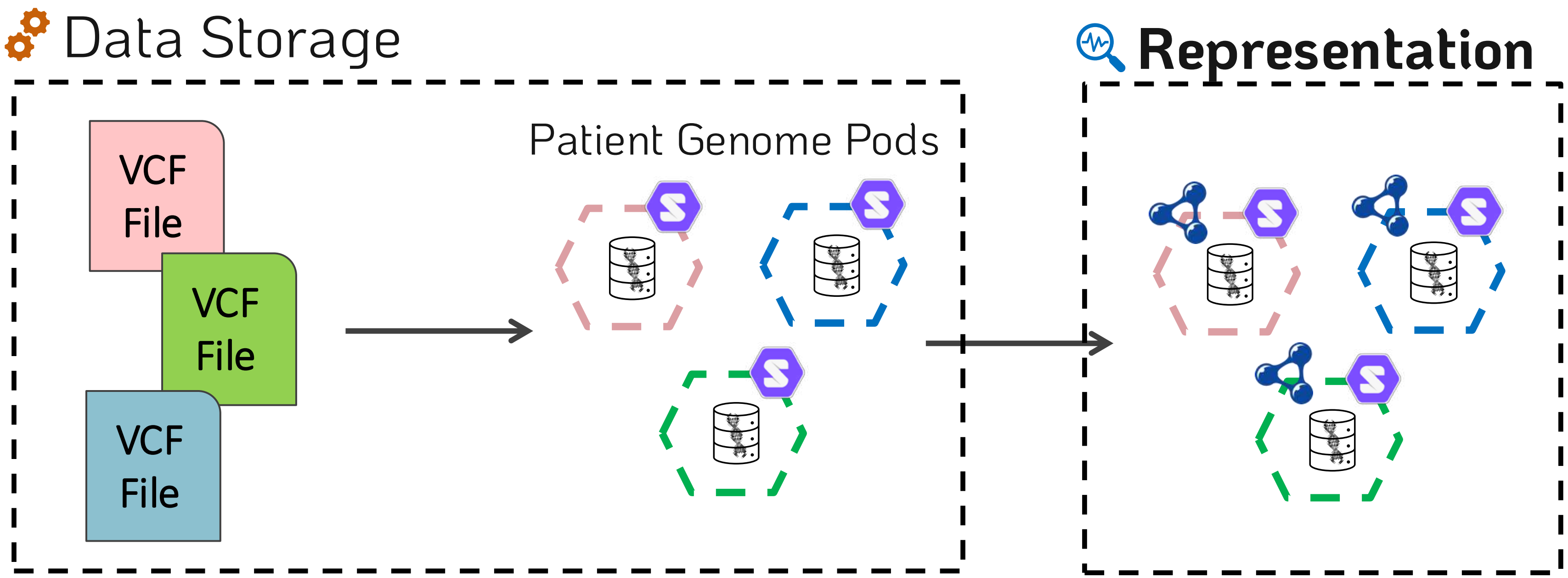
Solid can be implemented for genomic data storage.

Solid pod instances → Community Solid Server[1]

VCF Genomic Data → Illumina Platinum Genomes[2]

[1]https://github.com/CommunitySolidServer/CommunitySolidServer
[2]https://emea.illumina.com/platinumgenomes.html

Workflow Overview

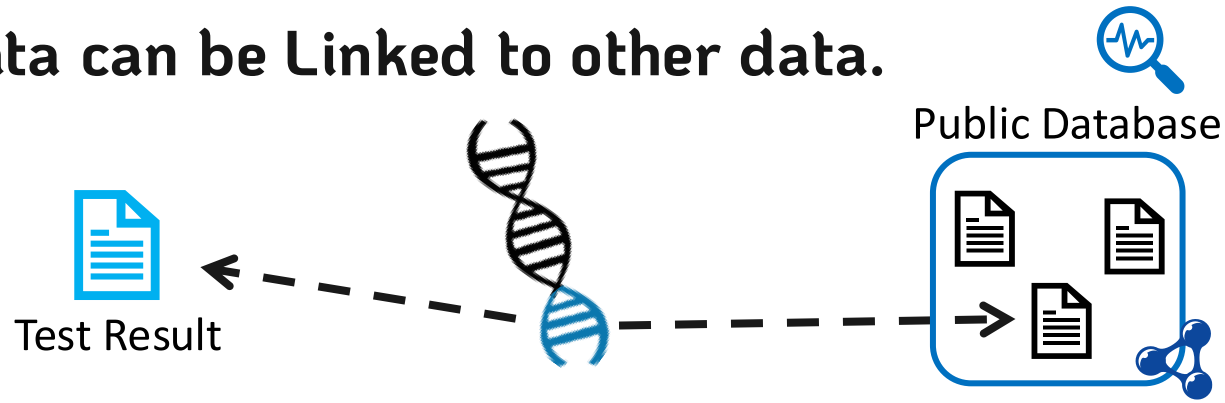


Genome Data in RDF

VCF genomic data can be represented as RDF.

VCF to RDF Vocabularies exist ⟶ SPHN Semantic Interoperability Framework[4]
+ FHIR HL7 Format[5]

VCF genomic data can be Linked to other data.

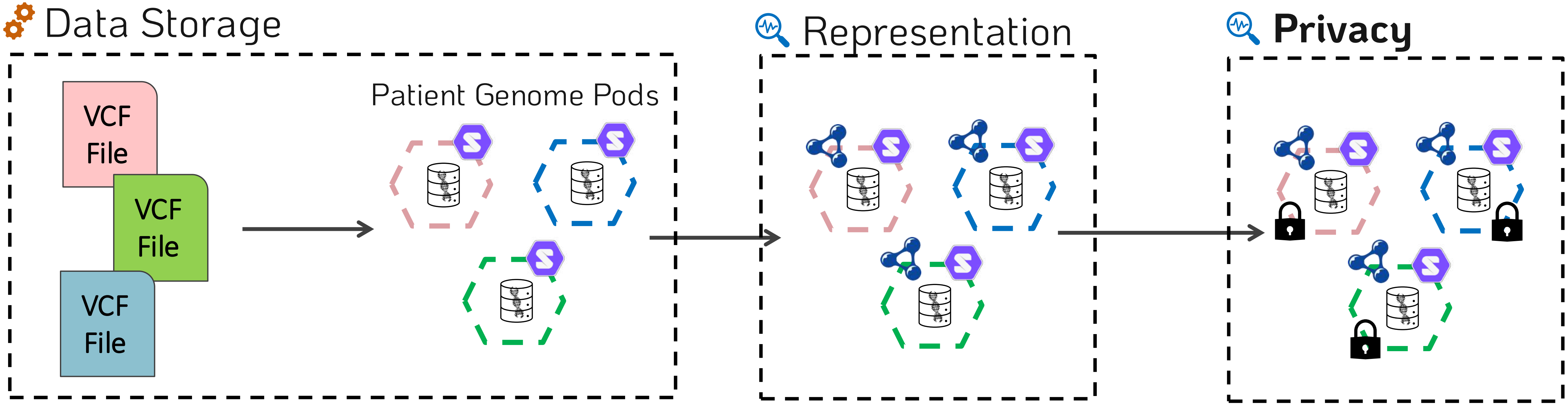


How to automate this?
What vocabularies could be used for the linkages?

[3] van der Horst E, et al. Bridging Clinical and Genomic Knowledge: An Extension of the SPHN RDF Schema for Seamless Integration and FAIRification of Omics Data. 2023.

[4] Dolin RH, et al. vcf2fhir: a utility to convert VCF files into HL7 FHIR format for genomics-EHR integration. 2021.

Workflow Overview



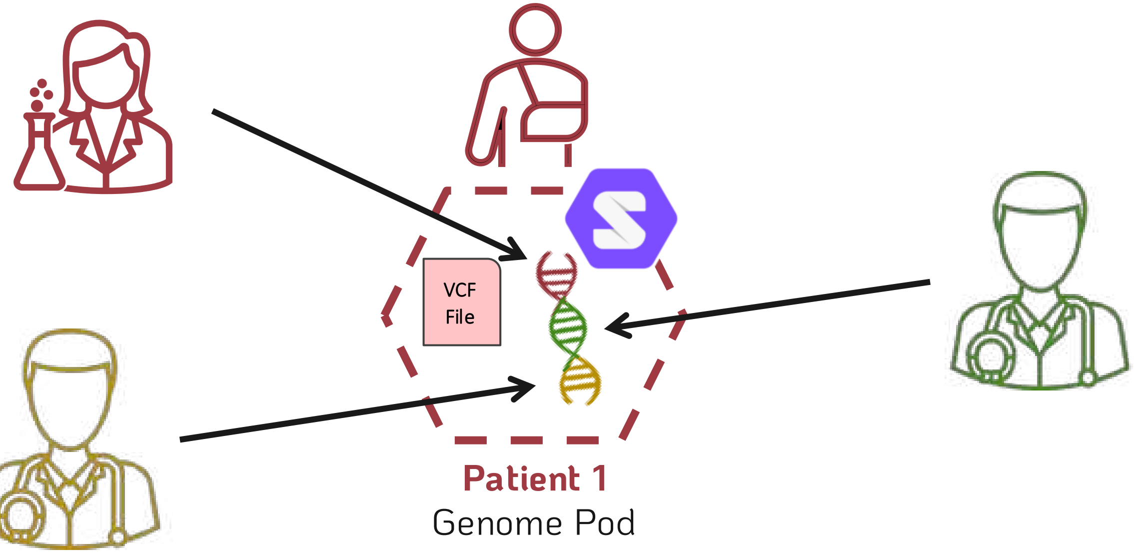
Data Privacy Controls

Granular privacy controls can be implemented in Solid for genomic data.

Web Access Contol → Solid Protocol ACP[3]

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

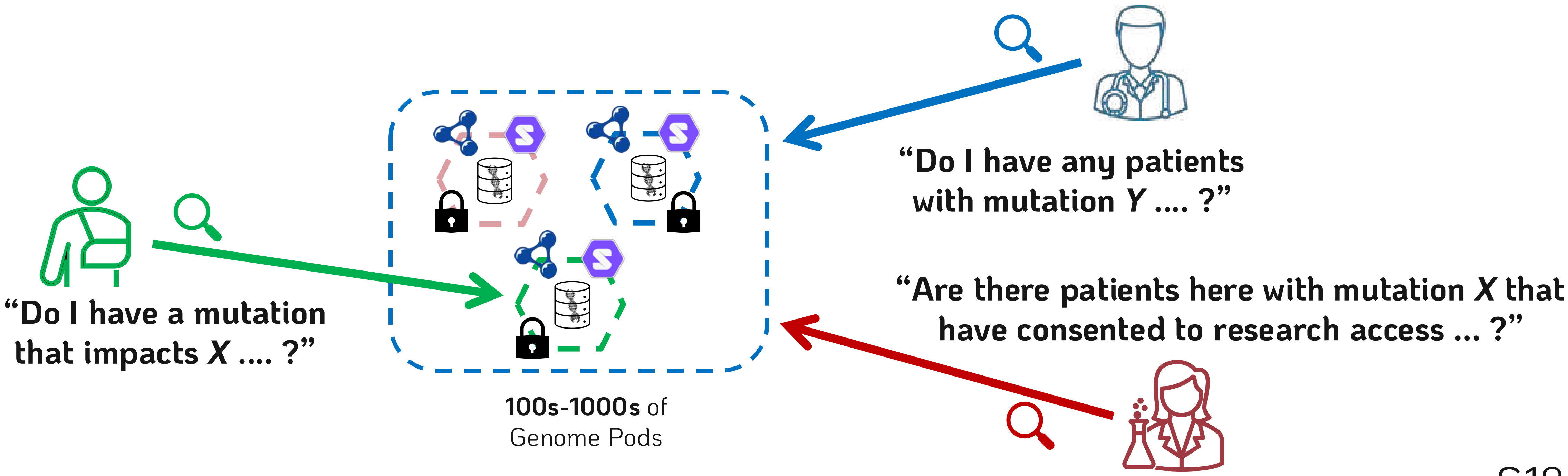
Mutation is also a semantic Class, above a triple but below a Chromosome

[5] https://solidproject.org/TR/protocol

Querying Genome Pods

RDF genomic data pods can be queried.

COMUNICA Query Engine → Custom Comunica implementation/engine[6]



[6] https://comunica.dev/

Querying Genome Pods (possibilities)

Single Pod Query

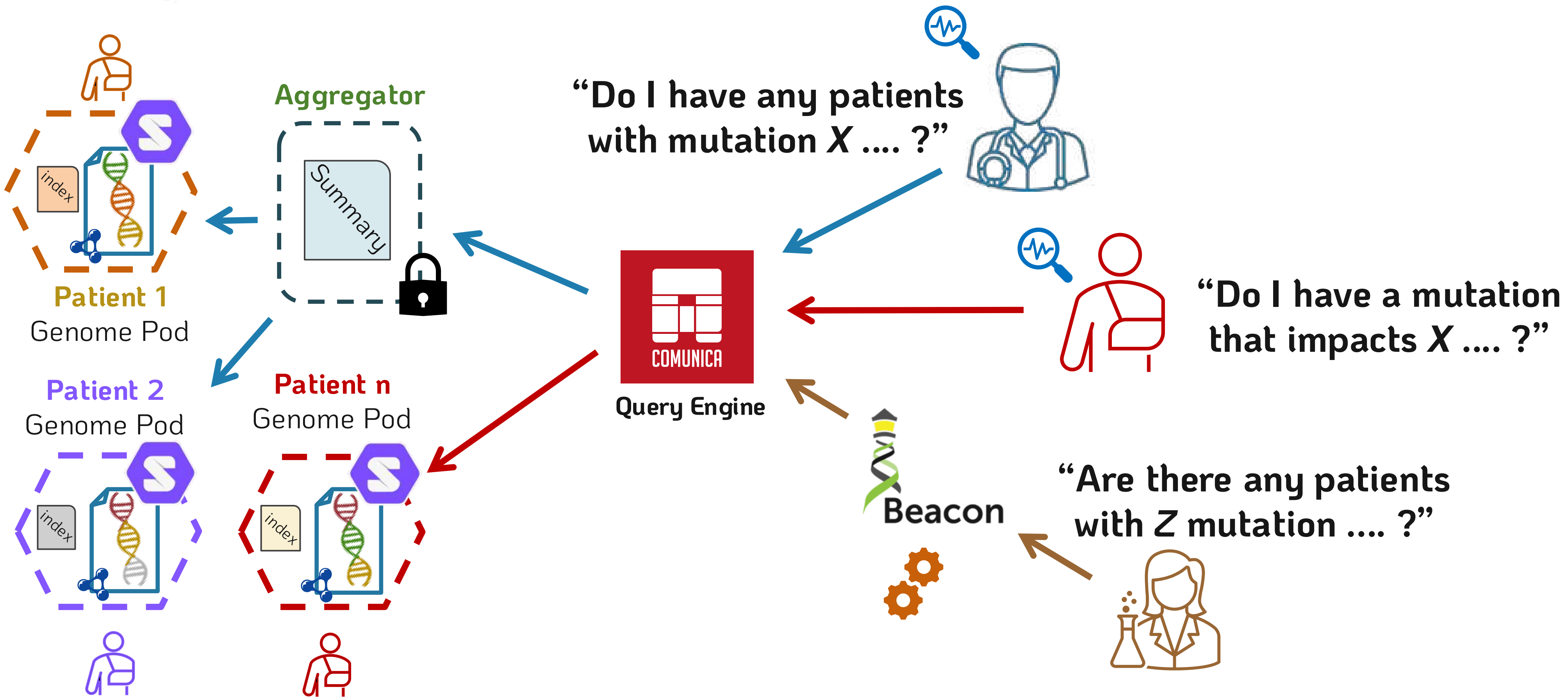
Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed



Implementation Specifics

Solid pod instances → Community Solid Server[1]

Genomic Data → Publicly available *VCF* data[2]

RDF Conversion → HERO Genomics + SPHN ontology[4]

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica[6]